

Dr. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech.

Branch : Petrochemical Engineering

Semester :VI

Subject Code & Name: BCHC602_Y21 Chemical Reaction Engineering - II

Max Marks: 60

Date: 20/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q.1	Objective type questions. (Compulsory Question)		12
1	For the irreversible elementary reaction in parallel: $A \xrightarrow{k_1} R$; $A \xrightarrow{k_2} S$ the rate of disappearance of reactant A is given by a. $(k_1 - k_2)C_A$ b. $(k_1 + k_2)C_A$ c. $0.5(k_1 + k_2)C_A$ d. k_1C_A	Level 1,2	1
2	A non-catalytic chemical reaction of the type $A \xrightarrow{+Z, k_1} B \xrightarrow{+Z, k_2} C \xrightarrow{+Z, k_3} D$ is called ----- --- reaction a. parallel b. series c. series-parallel d. None of these	Level 1,2	1
3	Enzymes (a protein) are catalysts found in organisms. Its efficiency of catalyzing a reaction is due to its capacity to lower the activation energy of the reaction. The enzyme ptyalin used for food digestion is present in a. Blood b. Saliva c. Intestine d. Gland	Level 1	1
4	Enzymes are destroyed, when the a. temperature is very high b. reactant's concentration is very high c. reactant's concentration is very low d. reaction rate is independent of the reactant's concentration	Level 1	1
5	Effective diffusivity (D_E) in a catalyst pellet is related to molecular diffusivity (D_N) and Knudsen diffusivity (D_K) as a. $D_E = D_M + D_K$ b. $\frac{1}{D_E} = \frac{1}{D_N} + \frac{1}{D_K}$ c. $D_E = D_M \cdot D_E$ d. $D_E = D_N/D_E$	Level 1,2	1
6	'If the catalyst pore size is small in comparison with the mean free path, collisions with the pore wall controls the process'. The diffusivity under this condition is called 'Knudsen diffusivity', which is affected by the a. Pressure b. temperature c. both (a) and (b) d. neither (a) nor (b)	Level 1,2	1
7	For multiple reactions, the flow pattern within the vessel affects the a. size requirement b. distribution of reaction products c. both (a) and (b) d. neither (a) nor (b)	Level 1	1
8	$A \xrightarrow{k_1} R, r_R = k_1 \cdot C_A^2; E_1 = 80 \text{ KJ/mole}$ $A \xrightarrow{k_2} S, r_S = k_2 \cdot C_A; E_2 = 120 \text{ KJ/mole}$ The desired product is R. A higher selectivity of R will be achieved, if the reaction is conducted at a. low temperature in a CSTR b. High temperature in a PFR c. low temperature in a PFR d. High temperature in a CSTR	Level 1,2	1
9	A pulse tracer is introduced in an ideal CSTR (with a mean residence time τ) at time, $t =$	Level 1,2	1

	0. The time taken for the exit concentration of the tracer to reach half of its initial value will be		
	a.) 2τ b. 0.5τ c. $\tau / 0.693$ d. 0.693τ		
10	Signal normally used to study non-ideal flow by stimulus response technique is _____ input	1,2	1
	a. pulse b. step c. both (a) and (b) d. neither (a) nor (b)		
11	The exit age distribution of fluid leaving a vessel is used to know the	1	1
	a. activation energies of a reaction b. reaction mechanism c. extent of non-ideal flow in the vessel d. none of these		
12	For a tubular reactor with space time ' τ ' and residence time ' θ ', the following statement holds good	1,2	1
	a. τ and θ are always equal. b. $\tau = \theta$, when the fluid density changes in the reactor. c. $\tau = \theta$, for an isothermal tubular reactor in which the density of the process fluid is constant. d. $\tau = \theta$, for a non-isothermal reactor.		
Q.2	Solve the following.		12
A)	For parallel reactions $A \xrightarrow{k_D} D$ (desired) $r_D = k_D C_A^{\alpha_1}$ $A \xrightarrow{k_U} U$ (undesired) $r_U = k_U C_A^{\alpha_2}$ Discuss all possible combinations to select the reaction scheme that will maximize $S_{D/U}$.	Level 1,2,3	6
B)	Derive the expression for C_{Rmax} , if the reaction $A \xrightarrow{k_1} R \xrightarrow{k_2} S$ is carried out in PFR.	Level 1,2,3	6
Q.3	Solve the following.		12
A)	For the following reaction mechanism $E + S \xrightarrow{k_1} E S$ $E S \xrightarrow{k_2} E + S$ $E S + W \xrightarrow{k_3} P + E$ show that $-r_s = \frac{V_{max} (S)}{K_m + (S)}$	Level 2,3	6
B)	The pyrolysis of acetaldehyde is believed to take place according to the following sequence: $\begin{array}{lcl} CH_3CHO & \xrightarrow{k_1} & CH_3^* + CHO^* \\ CH_3^* + CH_3CHO & \xrightarrow{k_2} & CH_3^* + CO + CH_4 \\ CHO^* + CH_3CHO & \xrightarrow{k_3} & CH_3^* + 2CO + H_2 \\ 2CH_3^* & \xrightarrow{k_4} & C_2H_6 \end{array}$ Derive the rate expression for the rate of disappearance of acetaldehyde, $-r_{Ac}$.	Level 2,3	6
Q.4	Solve Any Two of the following.		12
A)	Calculate the mass flux of reactant A to a single catalyst pellet 1 cm in diameter suspended in a large body of liquid. The reactant is present in dilute concentration, and the reaction is considered to take place instantaneously at the external pellet surface (i.e. $C_{AS} = 0$). The bulk concentration of reactant is 1.0 M, and the free system liquid velocity is 0.1m/s. the kinematic viscosity is 0.5 centistoke and the liquid diffusivity of A is 10^{-10}	Leve 2,3,4	6

$$\text{m}^2/\text{s}. T = 300 \text{ K}.$$

B) Discuss different types of boundary conditions used for diffusion in heterogeneous reactions.

Level 1,2

6

C) A plant is removing a trace of Cl_2 from a waste gas stream by passing it over a solid granular absorbent in a tubular packed bed (fig.). At present, 63.2% removal is being accomplished, but it is believed that greater removal could be achieved if the flow rate were increased by a factor of 4, the particle diameter were decreased by a factor of 3, and the packed tube length increased by 50%. What percentage of chlorine would be removed under the proposed scheme? (The chlorine transferring to the absorbent is removed completely by a virtually instantaneous chemical reaction).

Level 2,3,4

6

Q.5 Solve Any Two of the following.

12

A) Give brief account of mean residence time, variance and skewness used in study of in residence time distribution of chemical reaction.

Level 2,3

6

B) Twice in the last 25 years, gasoline shortages in the United States have produced long lines of motorists at service stations. The table below shows a distribution of the times required to obtain gasoline at 23 Centre County service stations. What is the average time required? If you were to ask randomly among those people waiting in line, "How long have you been waiting?" What would be the average of their answers?

Level 2,3,4

6

Total waiting Time (min)	0	3	6	9	12	15	18	21
Number of stations Having That Total Waiting Time	0	4	3	5	8	2	1	0

C) Give the informative account of maximum mixedness model.

Level 1,2

6

Q.6 Solve Any Two of the following.

12

A) The first – order reaction

$$A \longrightarrow B$$

is carried out in a 10-cm-diameter tubular reactor 6.36 m in length. The specific reaction rate is 0.25 min^{-1} . The results of a tracer test carried out on this reactor are shown in table.

Effluent Tracer Concentration as a Function of Time

t (min)	0	2	4	6	8	10	12	14	15	16	18	20
C (mg/L)	0	0	0.8	2	4.4	12.6	30	36	38	39	40	40

Calculate the conversion using the closed vessel dispersion model.

B) The following data were obtained from a tracer test to a reactor

Level 2,3,4

6

t (s)	0	5	10	15	20	25	30	35
C(t) (mg/dm ³)	0	0	0	5	10	5	0	0

i) Plot C and E curve.

ii) Find the fraction of material that spends between 15 and 20 seconds in the reactor.

C) Give brief account of one parameter model.

Level 1,2

6

END